PROG6212 POE PART 1

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MUSA MAPHANGA

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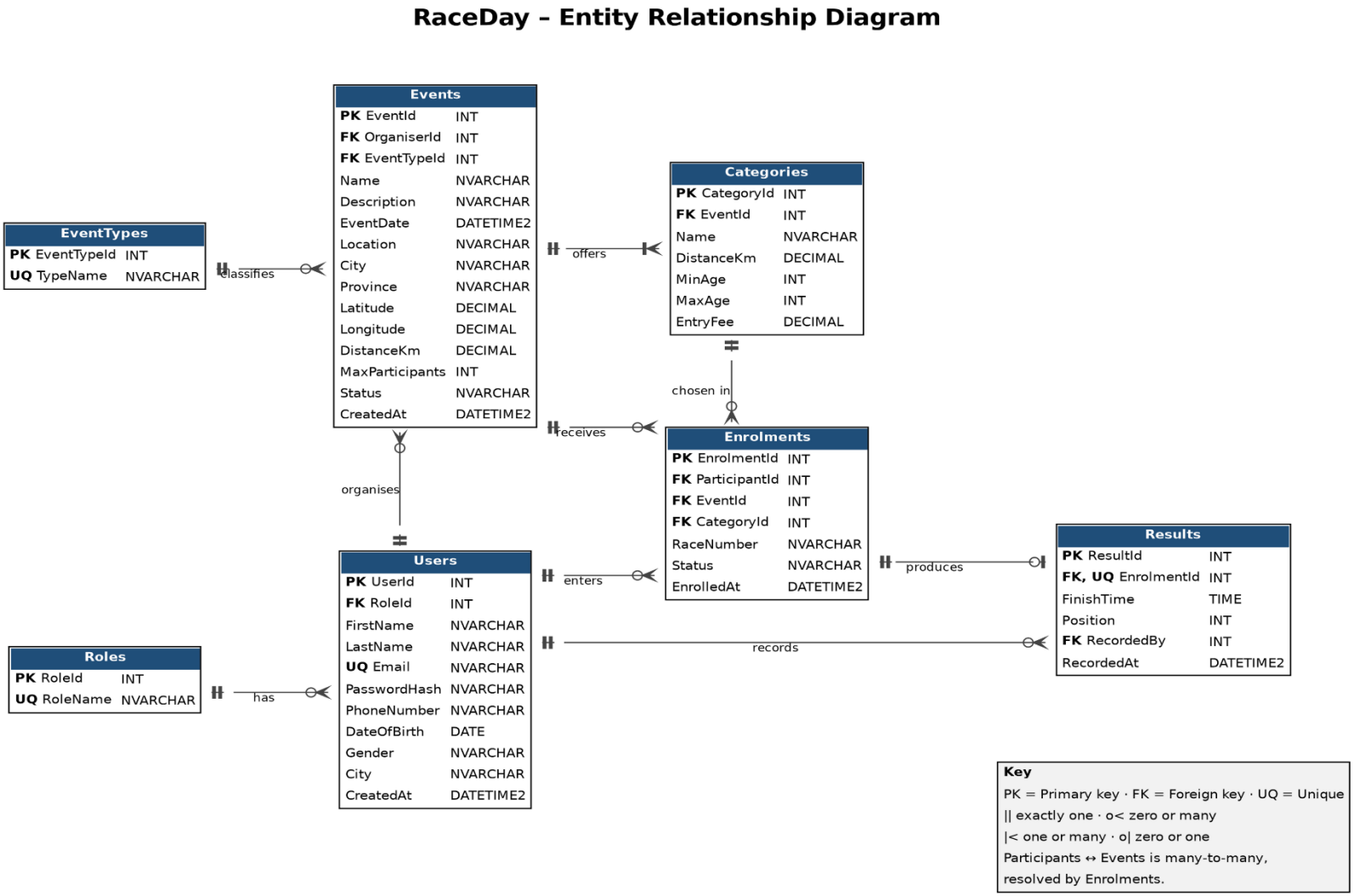
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Section A – Entity Relationship Diagram



Section B – API Endpoint Plan

RaceDay – API Endpoint Plan

This table lists all the endpoints I plan to build for the RaceDay API in Part 2. I used the Functional Requirements as a checklist, so every requirement is covered by at least one endpoint. I also added a few extra endpoints that I think the system will need, like changing a password, getting the list of event types, and showing the weather for an event.

How login and roles work

- When a user logs in, they get a JWT token. The token stores their user ID and their role (Organiser or Participant).
- "Owner" means the organiser who created that event. Only the owner can change their own events, categories, enrolments and results.
- If a protected endpoint is called without a valid token, it returns **401 Unauthorized**. If the user has the wrong role, it returns **403 Forbidden**.
- If the data sent is invalid, the endpoint returns **400 Bad Request** with a list of what is wrong

1. Authentication

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Lets a new user sign up as either an Organiser or a Participant.	None (public)	{ firstName, lastName, email, password, role, phoneNumber, dateOfBirth, gender, city }	201 Created – the new user's details (without the password). 400 Bad Request – missing or invalid fields. 409 Conflict – the email is already registered.
POST	/api/auth/login	Checks the user's email and password and gives back a token so they can use the system.	None (public)	{ email, password }	200 OK – { token, expiresAt, userId, role }. 401 Unauthorized – the email or password is wrong.

2. User Profile

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/users/me	Shows the profile of the user who is logged in.	Any (logged in)	None	200 OK – the user's profile. 404 Not Found – the user no longer exists.

PUT	/api/users/me	Lets the logged-in user update their own profile details.	Any (logged in)	{ firstName, lastName, phoneNumber, dateOfBirth, gender, city }	200 OK – the updated profile. 400 Bad Request – invalid fields.
PUT	/api/users/me/password	Lets the user change their password, but only if they enter their current password first.	Any (logged in)	{ currentPassword, newPassword }	204 No Content – password changed. 400 Bad Request – the new password is too weak. 401 Unauthorized – the current password is wrong.

3. Event Types

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/event-types	Gets the list of event types (Run, Walk, Cycle) so the front end can show them in a dropdown.	None (public)	None	200 OK – list of event types.

4. Events

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events	Lists events so users can browse them, with optional filters like ?type=&city=&province=&from=&to=&status=.	Any (logged in)	None	200 OK – list of events.

GET	/api/events/{id}	Shows all the details of one event, including its categories.	Any (logged in)	None	200 OK – the event and its categories. 404 Not Found – the event does not exist.
GET	/api/users/me/events	Lists only the events that the logged-in organiser created, for their dashboard.	Organiser	None	200 OK – list of the organiser's events.
POST	/api/events	Creates a new event, and the logged-in organiser becomes its owner.	Organiser	{ name, description, eventDate, location, city, province, latitude, longitude, distanceKm, eventTypeId, maxParticipants }	201 Created – the new event. 400 Bad Request – invalid fields or the date is in the past. 404 Not Found – the event type does not exist.

PUT	/api/events/{id}	Updates an event's details or changes its status (for example to Cancelled).	Organiser (owner)	{ name, description, eventDate, location, city, province, latitude, longitude, distanceKm, eventTypeId, maxParticipants, status }	200 OK – the updated event. 400 Bad Request – invalid fields. 403 Forbidden – not the owner. 404 Not Found – the event does not exist.
DELETE	/api/events/{id}	Deletes an event, but only if nobody has entered it yet (if people have entered, the organiser should cancel it instead).	Organiser (owner)	None	204 No Content – event deleted. 403 Forbidden – not the owner. 404 Not Found – the event does not exist. 409 Conflict – the event already

					has enrolments.
GET	/api/events/{id}/weather	Gets the weather forecast for the event's location on race day from an external weather API.	Any (logged in)	None	200 OK – weather forecast. 404 Not Found – the event does not exist. 502 Bad Gateway – the weather service is down.

5. Categories

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
GET	/api/events/{eventId}/categories	Lists the categories (like Under 20, Senior or 21km) that are available for an event.	Any (logged in)	None	200 OK – list of categories. 404 Not Found – the event does not exist.

POST	/api/events/{eventId}/categories	Adds a new age or distance category to an event.	Organiser (owner)	{ name, distanceKm, minAge, maxAge, entryFee }	201 Created – the new category. 400 Bad Request – invalid fields or minAge is bigger than maxAge. 403 Forbidden – not the owner. 404 Not Found – the event does not exist. 409 Conflict – this event already has a category with that name.
PUT	/api/categories/{id}	Updates the details of a category.	Organiser (owner)	{ name, distanceKm, minAge, maxAge, entryFee }	200 OK – the updated category. 400 Bad Request – invalid fields. 403 Forbidden – not the owner. 404 Not Found – the category does not exist. 409 Conflict – duplicate category name.
DELETE	/api/categories/{id}	Deletes a category, but only if no one has entered it yet.	Organiser (owner)	None	204 No Content – category deleted. 403 Forbidden – not the owner. 404 Not Found – the category does not exist. 409 Conflict – people have already entered this category.

6. Event Enrolments

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/events/{eventId}/enrolments	Enters the logged-in participant into an event under the category they chose, and saves the link between the participant, the event and the category.	Participant	{ categoryId }	201 Created – the enrolment with a race number. 400 Bad Request – the category is not part of this event, the participant's age doesn't fit the category, or the event is not open. 404 Not Found – the event or category does not exist. 409 Conflict – already entered, or the event is full.
GET	/api/users/me/enrolments	Lists all the events the logged-in participant has entered.	Participant	None	200 OK – list of enrolments with the event and category details.
DELETE	/api/enrolments/{id}	Lets a participant withdraw from an event before it happens (the status changes to Withdrawn instead of deleting the row).	Participant (own enrolment)	None	204 No Content – withdrawn. 403 Forbidden – this is not their enrolment. 404 Not Found – the enrolment does not exist. 409 Conflict – the event has already happened.

GET	/api/events/{eventId}/enrolments	Lists everyone who has entered one of the organiser's events.	Organiser (owner)	None	200 OK – list of enrolments with participant details. 403 Forbidden – not the owner. 404 Not Found – the event does not exist.

7. Results

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/enrolments/{enrolmentId}/result	Saves a participant's finish time and position after the event is over.	Organiser (owner)	{ finishTime, position }	201 Created – the saved result. 400 Bad Request – invalid time or position, or the event hasn't happened yet. 403 Forbidden – not the owner. 404 Not Found – the enrolment does not exist. 409 Conflict – a result was already saved for this enrolment.
PUT	/api/results/{id}	Fixes a result that was captured incorrectly.	Organiser (owner)	{ finishTime, position }	200 OK – the updated result. 400 Bad Request – invalid fields. 403 Forbidden – not the owner. 404 Not Found – the result does not exist.
GET	/api/events/{eventId}/results	Lists all the results for an event, sorted by position.	Organiser (owner)	None	200 OK – list of results. 403 Forbidden – not the owner. 404

					Not Found – the event does not exist.
GET	/api/users/me/results	Shows the logged-in participant's own results so they can track their performance over time.	Participant	None	200 OK – list of their results with the event, category, time and position.

Section C – SQL

